

SPRINT: Scalable and Predictive Intent Refinement for LLM-Enhanced Session-based Recommendation

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Abstract

Large language models (LLMs) have enhanced conventional recommendation models via user profiling, which generates representative textual profiles from users' historical interactions. However, their direct application to session-based recommendation (SBR) remains challenging due to severe session context scarcity and poor scalability. In this paper, we propose SPRINT, a scalable SBR framework that incorporates reliable and informative intents while ensuring high efficiency in both training and inference. SPRINT constrains LLM-based profiling with a global intent pool and validates inferred intents based on recommendation performance to mitigate noise and hallucinations under limited context. To ensure scalability, LLMs are selectively invoked only for uncertain sessions during training, while a lightweight intent predictor generalizes intent prediction to all sessions without LLM dependency at inference time. Experiments on real-world datasets show that SPRINT consistently outperforms state-of-the-art methods while providing more explainable recommendations.¹

CCS Concepts

• Information systems → Recommender systems.

Keywords

Session-based Recommendation, LLM-based user profiling

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¹Our code is available at https://github.com/Gyu-Seok0/SPRINT_SIGIR26.



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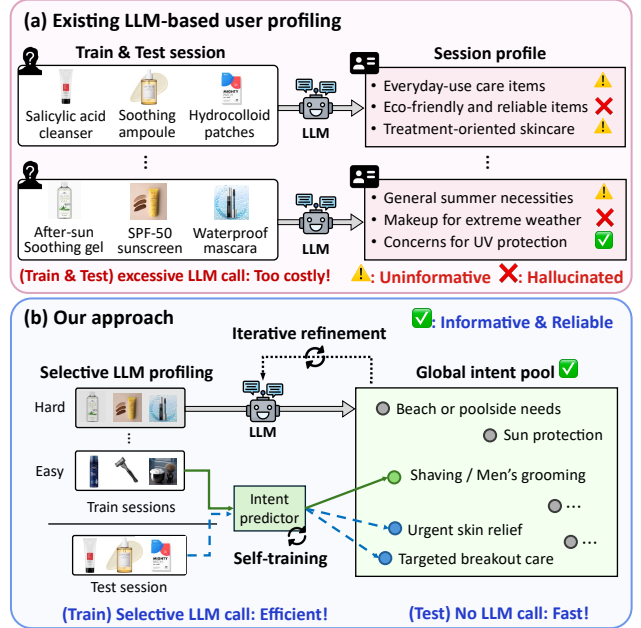


Figure 1: A conceptual comparison of (a) existing LLM-based user profiling and (b) our approach. Best viewed in color.

1 Introduction

With extensive world knowledge and reasoning capabilities, large language models (LLMs) have significantly enhanced conventional recommendation models (CRMs) [8, 72]. Through a deep understanding of natural language (e.g., user reviews, item descriptions), LLMs provide high-quality semantic information that is not directly captured by raw user-item interaction data [16, 20, 26, 63]. Existing work has incorporated this knowledge through various strategies, including embedding transfer [13, 32], semantic feature augmentation [29, 41], and auxiliary supervision [55, 58]. As a result, CRMs can leverage rich semantic knowledge to achieve substantially more accurate recommendations, while remaining highly efficient by avoiding costly LLM calls at inference time [35, 64].

A prominent line of research is LLM-based *user profiling*, which focuses on generating representative and descriptive textual profiles

from users’ historical interaction data [33, 38, 44, 47]. Since users’ preferences are only implicitly expressed through their behavioral histories, LLMs are leveraged for their knowledge and reasoning capabilities to unveil such latent preferences and synthesize them into explicit high-level descriptions, such as user profiles [38, 47], personas [49, 56], or intents [31, 44]. These profiles supply CRMs with explicit information about users’ underlying preferences that are challenging for models such as SASRec [18] to uncover solely from behavioral sequences.

However, directly applying existing LLM-based user profiling to session-based recommendation (SBR) is non-trivial. SBR is designed for practical scenarios, as many users browse online services without logging in or leaving persistent profiles [5, 12]. These design constraints shape the unique characteristics of session data: it is typically short and anonymous, preventing the use of user priors (e.g., metadata and historical records). Moreover, the absence of user IDs makes it impossible to determine which sessions belong to the same user, forcing each session to be processed independently and resulting in a massive volume of isolated sessions. Consequently, directly applying LLM-based user profiling to SBR faces two critical challenges: (C1) context scarcity and (C2) scalability issues.

(C1) Uninformative profiles due to session context scarcity:

Compared to traditional RS, SBR suffers from a more severe paucity of information [6, 44], which renders most existing LLM-based profiling methods ineffective. Specifically, each session contains only a small number of interactions (e.g., the average session length of Yelp is 3.63 [51]), making it difficult to generate informative user profiles. Moreover, the anonymous nature of sessions hinders the capture of user traits and long-term preferences [10, 46]. In this context-deficient environment, existing methods relying on substantial textual input and previous records are prone to noise and hallucinations, leading to misleading user behavior predictions. While a recent profile-based method [38] adopts self-reflection [15, 50] to ensure that profiles are grounded in the input information, it primarily targets logical consistency rather than capturing user preferences. Consequently, the generated profiles do not necessarily guarantee recommendation utility.

(C2) Scalability issues in SBR training and inference: Existing LLM-based profiling approaches suffer from severe scalability issues in both training and inference when applied to the SBR setting. During training, generating profiles for a massive number of sessions requires excessive LLM calls, making the offline pipeline significantly time-consuming and costly. Furthermore, the anonymity of sessions forces the system to generate profiles on-the-fly with LLMs during inference. This is further exacerbated by methods with advanced reasoning mechanisms such as iterative self-reflection [38], memory-based reflection [1] or prompt optimization [53], which may result in more than a doubling of training and inference time. Such substantial overhead renders existing approaches impractical for real-world SBR deployment, highlighting the need for a more scalable generation strategy.

In this paper, we propose **SPRINT**, a framework that enables **Scalable and PRedictive INTent** refinement for LLM-enhanced SBR (Figure 1). Specifically, SPRINT consists of two stages:

(1) In the first stage, we infer the underlying intents of sessions reliably and efficiently using LLMs. Since unconstrained free-form generation is prone to noise and hallucinations, we reformulate

intent generation as an *intent identification* problem over an expandable global intent pool. This pool serves as a shared and bounded intent space across sessions. To populate the intent pool in a scalable manner, we *selectively invoke* LLMs only for sessions where the SBR model exhibits high uncertainty, indicating insufficient contextual understanding. For such ambiguous cases, the inferred intents are refined through a predict-and-correct loop that iteratively validates their reliability based on their contribution to recommendation performance, ensuring their predictive utility.

(2) In the second stage, the generated intents are utilized to enhance the SBR model via an *intent predictor*. The intent predictor is a lightweight module that infers multiple intents per session without LLM dependency and can be flexibly integrated with existing SBR models. It is trained via a self-training strategy. Specifically, LLM-generated intents provide initial supervision, while confident predictions on sessions without LLM-derived intents are iteratively incorporated as pseudo-labels based on their interaction patterns. To further improve prediction accuracy, we devise a *collaborative intent enrichment* process, in which the predictor refines its predictions by leveraging neighboring sessions with similar behavioral patterns.

SPRINT systematically addresses the aforementioned challenges. For **C1**, SPRINT mitigates context scarcity by constraining LLM generation, validating inferred intents, and enriching them through collaborative signals across similar sessions. For **C2**, SPRINT resolves scalability issues through uncertainty-aware LLM invocation during training and lightweight, LLM-free intent prediction at inference time. Our core contributions are as follows:

- We identify two fundamental challenges in applying LLM-based user profiling to SBR that remain underexplored in the literature, highlighting the need for systematic investigation.
- We propose SPRINT, a scalable SBR framework that generates reliable and informative intents via selective LLM usage during training and enables lightweight, LLM-free intent prediction at inference time.
- We validate the effectiveness of SPRINT with comprehensive experiments and provide in-depth analyses of each proposed component. SPRINT consistently outperforms state-of-the-art methods while achieving substantially higher efficiency.

2 Related Work

Session-based recommendation (SBR). SBR aims to predict the next item given an anonymous session by modeling short-term user behavior [34, 36, 69–71] and is widely adopted for delivering timely recommendations without user authentication [19, 34, 37]. Early RNN-based methods encode temporal dynamics via gated recurrent units [12, 27]. Subsequent GNN-based methods model complex item transitions by linking items across sessions in a unified graph [62, 65], and Transformer-based models leverage self-attention to capture relationships between distant items within a session [40, 66]. However, SBR models suffer from data sparsity due to the short and anonymous nature of sessions [5, 57]. With such limited context, conventional SBR models struggle to capture reliable sequential patterns, resulting in suboptimal recommendations. To address this issue, prior work [5, 36, 39, 67] has explored modeling strategies that enrich session representations beyond raw interaction sequences.

Intent-aware recommendation. Among these efforts, a prominent direction is to model user intent within a session [4, 30, 45, 59, 67]. Intent reveals the underlying preferences behind user choices and provides contextual signals beyond mere interaction logs. As intent has no ground-truth labels, existing intent modeling can be broadly categorized into two groups: (1) explicit intent modeling, which utilizes side information (e.g., categories, titles) as intent labels [28, 36, 44], and (2) implicit intent modeling, which infers latent intents via unsupervised techniques like clustering [4, 39, 45]. Despite these advances, explicit methods remain vulnerable to noisy or incomplete side information, while implicit methods suffer from limited explainability and reliance on heuristic clustering choices. These limitations motivate the use of LLMs as a new paradigm in intent modeling, leveraging pretrained language knowledge to generate more robust and interpretable intents [31, 44, 53, 60, 61].

LLM-based SBR with intent profiling. Recent LLM-based SBR methods adapt user profiling to anonymous sessions through session-level intent profiling, and generally follow two paradigms: (1) LLM-as-SBR models [53, 61] and (2) LLM-enhanced SBR [31, 44].

In the first paradigm, LLMs serve as standalone SBR models that directly infer user intents and rank items based on the intents. PO4ISR [53] optimizes prompts to extract user intents and then ranks candidates accordingly. Re2LLM [61] reinforces prompts by injecting hints to fix recommendation errors. Though effective, directly deploying LLMs for recommendation inevitably incurs substantial inference-time overhead, making them less preferred for real-time applications (see §5.3 for a detailed analysis).

In contrast, LLM-enhanced SBR methods employ LLMs to generate user intents that guide conventional SBR models. EGRec [31] jointly trains an LLM for intent generation with a recommender, while LLM4SBR [44] infers long- and short-term intents from item titles to enrich session representations. Despite their effectiveness, EGRec requires costly fine-tuning and risks degrading the LLM’s pretrained knowledge, while LLM4SBR limits intent expressiveness by treating intents as simple item titles. Moreover, both methods rely on LLM-generated outputs that are not explicitly optimized for recommendation performance. This can introduce spurious patterns and potentially degrade recommendation accuracy. These limitations highlight the need for an LLM-enhanced SBR approach that explicitly validates generated intents while remaining efficient and scalable for deployment.

3 Problem Formulation

3.1 Preliminaries

Notations. Let $S = [i_1, i_2, \dots, i_N]$ be a session of N items from the item set $\mathcal{I}$. Given S , the goal of SBR is to predict the next item i_{next} . The set of all sessions $\mathcal{S}$ is partitioned into disjoint training and test sets, denoted by $\mathcal{S}_{\text{train}}$ and $\mathcal{S}_{\text{test}}$, such that $\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset$. Let $\mathcal{F}$ be the set of feature fields for items (e.g., title, description, categories, reviews). For each item, we collect its features into a tuple $\mathbf{x}_i = (x_f)_{f \in \mathcal{F}}$, where x_f denotes the raw data for field f . Given a session S , we define the session feature matrix as $\mathbf{X}_S = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]$.

Conventional SBR. Let $\text{SBR}(\cdot)$ be a conventional SBR model (e.g., RNN- [12, 27] and Transformer-based [40, 66]). Given a session S ,

it estimates the probability distribution for the next item as follows:

$$P_\theta(i_{\text{next}} | S) \leftarrow \text{SBR}(S, \mathbf{X}_S), \quad (1)$$

where θ denotes the parameters of the SBR model. The next item is predicted as: $\hat{i}_{\text{next}} = \arg\max_{j \in \mathcal{I}} P_\theta(j | S)$.

3.2 Problem Definition

We study LLM-based intent profiling in the context of SBR, where LLMs are used to generate explicit session-level intents to enhance conventional SBR models. Let $\mathbf{Z}_S = \text{LLM}(\mathbf{X}_S)$ be the output from the LLM, which may take various forms such as profiles [38, 47], personas [49, 56], or intents [31, 44]. These signals are then provided alongside the session:

$$P_\theta(i_{\text{next}} | S) \leftarrow \text{SBR}(S, \mathbf{X}_S, \mathbf{Z}_S). \quad (2)$$

$\mathbf{Z}_S$ offers rich contextual information not directly observable from S , thereby enhancing the learning of SBR models. Despite its promise, LLM-based profiling for SBR is limited by two aforementioned challenges that remain insufficiently explored in prior work [33, 38, 44, 47]: (C1) uninformative profiles caused by session context scarcity, and (C2) scalability issues in SBR training and inference.

4 SPRINT

We propose SPRINT, a two-stage framework that enhances the SBR model with LLM-generated intents. In Stage 1, we generate session-level multiple intents by harnessing LLMs (§4.1). In Stage 2, we integrate these intents into the SBR model to enhance recommendation performance (§4.2). Finally, we describe the optimization process (§4.3). Figure 2 shows an overview of SPRINT.

4.1 Stage 1: Scalable and Predictive Intent Refinement via LLMs

The goal of this stage is to infer a set of session-level intents that provide informative signals for predicting user behavior in a scalable manner, which will be used to enhance SBR models. We first identify training sessions with high uncertainty using a conventional SBR model to selectively invoke LLM reasoning (§4.1.1). Then, we construct a global intent pool to serve as a shared and constrained space of intents, and apply an iterative predict-and-correct loop to obtain reliable and informative intents (§4.1.2).

4.1.1 Uncertainty-aware hard session selection. To reduce the substantial computational cost of generating LLM-based intents for all sessions (C2), we selectively allocate the LLM’s reasoning to the hard sessions via an uncertainty-aware selection strategy. This approach employs a base SBR model, warmed up exclusively on the training sessions, to identify sessions with high uncertainty. We prioritize these sessions—where underlying patterns are difficult to discern from raw interactions alone—as they stand to benefit most from the nuanced reasoning of LLMs.

For each training session S , we quantify its uncertainty via two metrics: session entropy $H_{\text{sess}}(S)$ and prediction difficulty $H_{\text{pred}}(S)$.

$$H_{\text{sess}}(S) = - \sum_{i \in S} P_\theta(i | S) \log P_\theta(i | S), \quad (3)$$

$$H_{\text{pred}}(S) = - \log P_\theta(i_N | S_{1:N-1}),$$

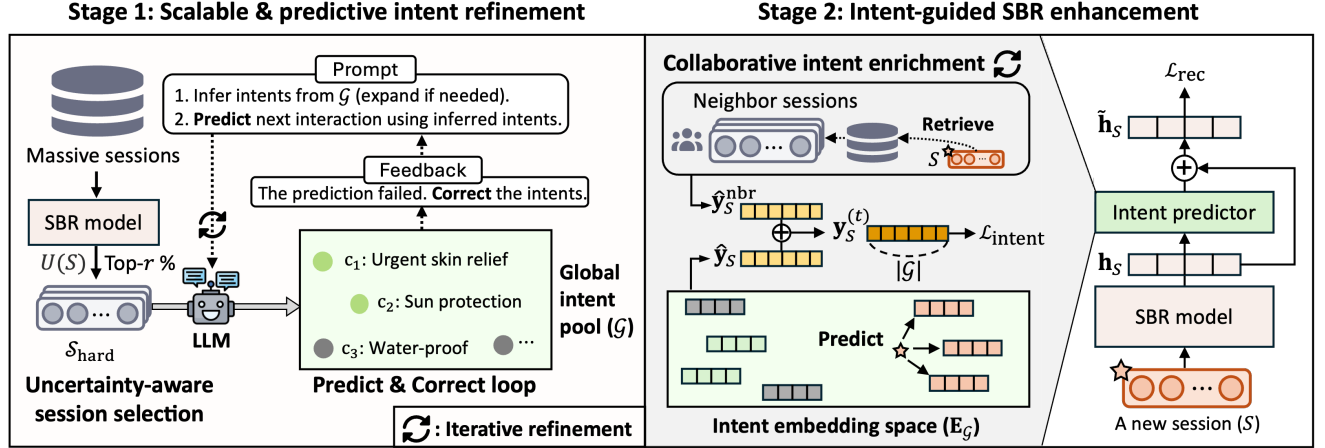


Figure 2: Overview of the SPRINT framework. Best viewed in color.

where $P_\theta(i|S) = \frac{\exp(\mathbf{h}_S^\top \mathbf{e}_i)}{\sum_{j \in I} \exp(\mathbf{h}_S^\top \mathbf{e}_j)}$, and i_N is the held-out last item of the session S . $\mathbf{h}_S \in \mathbb{R}^d$ is the session representation² and $\mathbf{e}_i \in \mathbb{R}^d$ is the embedding of item i . Specifically, $H_{\text{sess}}(S)$ reflects internal interest inconsistency, where high values indicate diverse item interactions within a session. In contrast, $H_{\text{pred}}(S)$ measures the model’s lack of confidence in predicting i_N .

Since $H_{\text{sess}}(S)$ and $H_{\text{pred}}(S)$ capture complementary aspects of session uncertainty, we combine them into a unified score $U(S)$ using reciprocal rank fusion [7, 22] for scale-invariant integration.

$$U(S) = \frac{1}{k + r_{\text{sess}}(S)} + \frac{1}{k + r_{\text{pred}}(S)}. \quad (4)$$

$r_{\text{sess}}(S)$ and $r_{\text{pred}}(S)$ denote the ranks of session S among all training sessions sorted in descending order of $H_{\text{sess}}(S)$ and $H_{\text{pred}}(S)$, respectively. We set the damping constant $k = 60$ following [3].

Using $U(S)$, we select the top- $r\%$ most ambiguous sessions, denoted as S_{hard} . In this work, we set $r = 10$. Then, the LLM infers intents exclusively for S_{hard} , thereby substantially improving the scalability of our framework. We show that our strategy achieves significantly reduced training time and robust performance across different values of r . We provide a detailed analysis in §5.3.

4.1.2 Predictive intent refinement. A straightforward way to acquire intents is to simply instruct LLMs to infer them for each session [31, 44, 53]. However, due to session context scarcity (C1), this approach has several limitations. First, it lacks control over the generation; there is always a risk of hallucination, and LLMs often use different terms to refer to the same concept (e.g., “eco-friendly” vs “environmentally friendly” skincare), making it difficult to handle the outputs consistently. Second, even when a generated intent appears reasonable, it often remains overly broad (e.g., “summer necessities”), providing limited insight into user behavior.

As a solution, we propose a novel approach to improve the reliability and informativeness of LLM-generated intents. We first construct the *Global Intent Pool*, a shared and constrained set of intents. We then use LLMs to pinpoint the most relevant intents

from the pool and expand it only when no appropriate candidates are available. By grounding intent generation in the intent pool, we constrain the output space of LLMs, reducing hallucination and improving consistency in expression. Furthermore, the generated intents are iteratively refined and validated through the *Predict-and-Correct Loop* to ensure the quality of the intents.

Global Intent Pool (GIP). We introduce GIP, denoted as $\mathcal{G}$, to constrain the LLM’s output space for session-level intent identification. We first initialize $\mathcal{G}$ by instructing the LLM using a prompt $\mathcal{P}_{\text{GIP}}$ conditioned on the recommendation domain D (e.g., Beauty): “Generate a minimal set of distinct intents from general to domain-specific for $\langle D \rangle$.” This process yields basic, domain-relevant concepts (e.g., “anti-aging”, “fragrance-free products”), formally defined as:

$$\mathcal{G} = \text{LLM}(\mathcal{P}_{\text{GIP}}(D)). \quad (5)$$

After initialization, the P&C loop is executed per session S : The LLM identifies relevant intents from $\mathcal{G}$ or generates a new intent c' only when no suitable candidates exist. c' is integrated into $\mathcal{G}$ only if it contributes to an accurate prediction of user behavior via $\mathcal{G} \leftarrow \mathcal{G} \cup \{c'\}$. By sharing predefined intents and generating new ones only as needed, GIP constrains the proliferation of redundant concepts, maintaining a compact and discriminative intent set.³

Predict-and-Correct (P&C) loop. We introduce the P&C loop, which iteratively refines and validates inferred intents to ensure high informativeness. We define intent informativeness through recommendation utility: an intent is valid only if it facilitates the accurate prediction of subsequent user behaviors.

For each session $S \in S_{\text{hard}}$, we first construct a small *intent validation task* by holding out the last item i_N from S . We then define a candidate set $I_S = \{i_N, i_{\text{out}_1}, \dots, i_{\text{out}_M}\}$, where $\{i_{\text{out}_m}\}_{m=1}^M$ are M out-of-session items not observed in S .⁴ The LLM is instructed to infer underlying intents from session S , and then **predict** the item $\hat{i} \in I_S$ that is most likely to be the held-out item based on the inferred intents. If $\hat{i}$ matches i_N , the inferred intents are accepted;

²SPRINT seamlessly integrates with various conventional SBR models that produce session representations. For instance, $\mathbf{h}_S$ corresponds to the hidden state of the last item in the session for both RNN-based [12, 27] and transformer-based models [18, 52].

³In our experiments, the final number of generated intents is 120, 135, and 118 for Beauty, Yelp, and Book, respectively.

⁴As the simplest choice, we randomly select the out-of-session items. M is set to 5.

Algorithm 1: P&C loop

Input : Hard session $S \in \mathcal{S}_{\text{hard}}$, GIP $\mathcal{G}$, max iterations T
Output : Final predictive intent set C_S^{LLM} , Updated GIP $\mathcal{G}$

```

1 Construct the candidate item set  $I_S$ 
2 Construct the initial LLM input  $V_S^{(0)} \leftarrow (X_S, X_{I_S}, \mathcal{G})$ 
3 Initialize the set of intents  $C_S^{\text{LLM}} \leftarrow \emptyset$ 
4 for  $t = 0$  to  $T - 1$  do
    // Infer intents and predict the held-out item
    5  $C_S^{(t)}, \hat{i} \leftarrow \text{LLM}(\mathcal{P}_{P\&C}(V_S^{(t)}))$  ▷ Eq. (6)
    6 if  $\hat{i} = i_N$  then
    7     Accept the final intent set  $C_S^{\text{LLM}} \leftarrow C_S^{(t)}$ 
    8     break
    9 else
    10  $R_S^{(t)} \leftarrow$  ("The prediction  $\hat{i}$  failed. Correct  $C_S^{(t)}$  and retry.")
    11 Update the LLM input  $V_S^{(t+1)} \leftarrow (V_S^{(t)}, R_S^{(t)})$  ▷ Eq. (7)
    // Update GIP with newly generated intents
12  $\mathcal{G} \leftarrow \mathcal{G} \cup C_S^{\text{LLM}}$ 
13 return  $C_S^{\text{LLM}}, \mathcal{G}$ 

```

otherwise, they are rejected. The LLM is then provided with verbal feedback to rectify its reasoning process and **correct** the intents.

Specifically, the initial iteration begins by obtaining the feature matrix for the candidate set, $X_{I_S} = \{\mathbf{x}_N, \mathbf{x}_{\text{out}_1}, \dots, \mathbf{x}_{\text{out}_M}\}$, to construct the starting input $V_S^{(0)} = (X_S, X_{I_S}, \mathcal{G})$. The LLM then jointly generates a set of session intents $C_S^{(0)}$ and predicts an item $\hat{i}$ via:

$$C_S^{(0)}, \hat{i} \leftarrow \text{LLM}(\mathcal{P}_{P\&C}(V_S^{(0)})), \quad (6)$$

where the two tasks in prompt $\mathcal{P}_{P\&C}$ are defined as follows.

$\mathcal{P}_{P\&C}$ = "(1) Infer multiple session-level intents $C_S^{(0)}$ from $\mathcal{G}$; generate novel and distinct intents only when no suitable candidate is identified. (2) Predict the held-out item from I_S using $C_S^{(0)}$."

If $\hat{i} = i_N$, we terminate the loop and accept $C_S^{(0)}$ as the final predictive intent set for session S , denoted as C_S^{LLM} . Any newly generated intents in C_S^{LLM} are added to $\mathcal{G}$. Otherwise, we append explicit verbal feedback $R_S^{(0)}$ (e.g., "The prediction $\hat{i}$ failed. Correct the intents $C_S^{(0)}$ and retry.") to the previous input:

$$V_S^{(1)} = (V_S^{(0)}, R_S^{(0)}). \quad (7)$$

This loop continues up to T iterations, and any sessions that remain inconclusive within T iterations are subsequently handled during training (§4.2.2). In this work, we set $T = 3$. The number of intents and iterations required for successful prediction varies across sessions and is automatically determined by the LLM based on the inherent complexity of each user's behavior. The overall procedure of the P&C loop is provided in Algorithm 1.

4.2 Stage 2: Intent-guided SBR Enhancement

This stage aims to enhance SBR models using LLM-generated intents. A naive approach is to treat them as static input features; for example, one may obtain their textual embeddings and feed them into the model. However, this is suboptimal for two reasons: (1) it

is inapplicable to sessions without pre-generated intents, necessitating costly LLM inference at test time; and (2) it fails to reflect the varying importance of multiple intents within a session.

We first introduce a multi-intent prediction to identify diverse user motivation (§4.2.1). We then present collaborative intent enrichment to effectively impute missing intent signals (§4.2.2). Finally, these multi-faceted intents are integrated into the SBR model through an intent fusion mechanism (§4.2.3).

4.2.1 Intent predictor. While intents are generated for hard sessions in $\mathcal{S}_{\text{hard}}$ during Stage 1, the discovered intent concepts in $\mathcal{G}$ can also be used to better understand other sessions. To avoid repetitive LLM calls, we train a lightweight intent predictor [17, 21] to generalize the LLM's reasoning from $\mathcal{S}_{\text{hard}}$ to remaining sessions in $\mathcal{S}_{\text{train}} \setminus \mathcal{S}_{\text{hard}}$. This enables scalable intent acquisition across the entire dataset and also avoids LLM dependency at test time (C2).

Let $\mathbf{E}_{\mathcal{G}} = \{\mathbf{e}_c\}_{c \in \mathcal{G}}$ be the set of intent embeddings, where $\mathbf{e}_c \in \mathbb{R}^d$ is the embedding of intent c . To model the session-intent relevance, we employ the query-key-value design [54], where the relevance is computed by matching a query and a key. Specifically, the session query $\mathbf{q}_S$, the intent key $\mathbf{k}_c$, and the value $\mathbf{v}_c$ are obtained as follows:

$$\mathbf{q}_S = \mathbf{W}_q \mathbf{h}_S \in \mathbb{R}^d, \quad \mathbf{k}_c = \mathbf{W}_k \mathbf{e}_c \in \mathbb{R}^d, \quad \mathbf{v}_c = \mathbf{W}_v \mathbf{e}_c \in \mathbb{R}^d, \quad (8)$$

where $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v \in \mathbb{R}^{d \times d}$ are the projection matrices for queries, keys, and values, respectively. We compute the session-intent relevance, denoted as $\hat{\mathbf{y}}_S = [\hat{y}_{Sc}]_{c \in \mathcal{G}}$, as follows:

$$\hat{y}_{Sc} = \sigma(\mathbf{q}_S^\top \mathbf{k}_c) \in [0, 1], \quad (9)$$

where σ denotes the sigmoid function [48]. Unlike the softmax function [2], which enforces the mutual exclusivity of intents via $\sum_c \hat{y}_{Sc} = 1$, the sigmoid activation provides more flexibility to capture multiple relevant intents non-exclusively. The session-intent relevance is then learned by the following loss:

$$\mathcal{L}_{\text{intent}} = -\frac{1}{|\mathcal{G}|} \sum_{c \in \mathcal{G}} \left[y_{Sc}^{(t)} \log \hat{y}_{Sc} + (1 - y_{Sc}^{(t)}) \log (1 - \hat{y}_{Sc}) \right], \quad (10)$$

where $\mathbf{y}_S^{(t)} = [y_{Sc}^{(t)}]_{c \in \mathcal{G}}$ denotes the intent labels at training epoch t . In the early stages of training, the labels are derived from LLM-generated intents $\{C_S^{\text{LLM}}\}_{S \in \mathcal{S}_{\text{hard}}}$. As training progresses, we incorporate enriched intent labels to cover all training sessions, a process further elaborated in the following subsection.

4.2.2 Collaborative intent enrichment. To accurately train the intent predictor, we propose a collaborative intent enrichment strategy. This approach iteratively refines intent supervision by leveraging neighboring sessions with similar behavioral patterns.

At the beginning of training, we assign initial intent labels $\mathbf{y}_S^{(0)} = [y_{Sc}^{(0)}]_{c \in \mathcal{G}}$ for sessions where LLM-derived intents are available.

$$y_{Sc}^{(0)} = \begin{cases} 1, & c \in C_S^{\text{LLM}}, \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

During training, we progressively enrich missing intent labels using predictions from the intent predictor (i.e., $\hat{\mathbf{y}}_S$ in Eq. (9)). This procedure is inspired by self-training [11, 43], a well-established semi-supervised learning paradigm where a model learns from its own predictions on unlabeled data.

However, predictions made for individual sessions can be unstable and noisy, particularly in the early stages of training. To improve robustness, we propagate intent information from behaviorally similar sessions. Specifically, given a session representation $\mathbf{h}_s$, we first retrieve its top- K most similar sessions within the same mini-batch $\mathcal{B}$ based on cosine similarity:

$$\mathcal{N}(S) = \text{TopK}(\{\text{sim}_{S'}\}_{S' \in \mathcal{B}, S' \neq S}), \quad \text{sim}_{S'} = \cos(\mathbf{h}_s, \mathbf{h}_{S'}). \quad (12)$$

We then aggregate neighbor intent predictions into $\hat{\mathbf{y}}_s^{\text{nbr}}$ using similarity weights $w_{S'}$ ⁵:

$$\hat{\mathbf{y}}_s^{\text{nbr}} = \sum_{S' \in \mathcal{N}(S)} w_{S'} \cdot \hat{\mathbf{y}}_{S'}. \quad (13)$$

At training epoch t , we obtain the enriched intent labels $\mathbf{y}_s^{(t)}$ by averaging the session-level and neighbor-aggregated predictions:

$$\mathbf{y}_s^{(t)} = \frac{1}{2}(\hat{\mathbf{y}}_s + \hat{\mathbf{y}}_s^{\text{nbr}}). \quad (14)$$

These enriched labels $\mathbf{y}_s^{(t)}$ are then used as targets for $\mathcal{L}_{\text{intent}}$ (Eq. (10)).

The enriched labels $\mathbf{y}_s^{(t)}$ provide richer supervision in two aspects. First, when explicit intent supervision is missing, they supply pseudo-intents inferred from behaviorally similar sessions. Notably, compared to relying solely on a single session-level prediction $\hat{\mathbf{y}}_s$, our neighborhood-based enrichment yields a more stable and reliable signal. Second, unlike the initial binary labels $\mathbf{y}_s^{(0)}$, the enriched labels $\mathbf{y}_s^{(t)}$ encode soft probabilities that reflect the importance of each intent for a given session, providing more nuanced supervision for learning. As updating the labels every epoch is both inefficient and unnecessary, we update $\mathbf{y}_s^{(t)}$ every ρ epochs. We set ρ to 5.

4.2.3 Fusing multi-intent information. We enhance the session representation with intents based on their varying importance to the session. We create the guidance representation $\mathbf{g}_s$ as follows:

$$\mathbf{g}_s = \text{Dropout}\left(\text{LayerNorm}\left(\mathbf{h}_s + \sum_{c \in \mathcal{G}} \phi(\hat{\mathbf{y}}_{sc}) \mathbf{v}_c\right)\right), \quad (15)$$

where $\phi(a) = a \cdot \mathbb{I}[a > \tau]$ is a filtering function with threshold τ . $\mathbf{v}_c$ is the intent-aspect value vector defined in Eq (8). This filtering step allows the model to focus on intents above a certain degree of relevance, while preserving their relative importance. In this work, we set $\tau = 0.5$. The final intent-enhanced representation is obtained as: $\tilde{\mathbf{h}}_s = \mathbf{h}_s + \mathbf{g}_s$. The standard next-item prediction loss is adopted:

$$\mathcal{L}_{\text{rec}} = - \sum_{S \in \mathcal{S}_{\text{train}}} \left[\log \sigma(\mathbf{e}_{i_{\text{pos}}}^\top \tilde{\mathbf{h}}_s) + \sum_{i_{\text{neg}}} \log (1 - \sigma(\mathbf{e}_{i_{\text{neg}}}^\top \tilde{\mathbf{h}}_s)) \right]. \quad (16)$$

For each training session, we sample two negative items: one randomly sampled negative and one hard negative selected from items mispredicted in the P&C loop. If no hard negatives are available, both negatives are sampled at random [23].

4.3 Optimization of SPRINT

The final learning objective of SPRINT is defined as:

$$\min_{\theta} \mathcal{L}_{\text{rec}} + \lambda_{\text{intent}} \mathcal{L}_{\text{intent}} + \lambda_{\text{ortho}} \mathcal{L}_{\text{ortho}}, \quad (17)$$

where $\theta = \{\text{SBR}(\cdot), \mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v, \mathbf{E}_g\}$ denotes the parameters of SPRINT. We additionally adopt an intent-orthogonality loss, which

⁵ $w_{S'} = \exp(\text{sim}_{S'}) / \sum_{S'' \in \mathcal{N}(S)} \exp(\text{sim}_{S''})$

Table 1: Dataset statistics after preprocessing.

Dataset		#Sessions	#Items	#Interactions	Avg. Length	Metadata
Beauty	Total	7,524	2,351	31,538	4.19	description, categories, rating, title, review, price
	Train	6,448	2,351	27,130	4.21	
	Valid	611	1,126	2,480	4.10	
	Test	465	1,085	1,928	4.15	
Yelp	Total	41,892	8,701	152,818	3.65	name, stars, categories, text
	Train	34,296	8,701	125,181	3.65	
	Valid	3,985	5,112	14,453	3.63	
	Test	3,611	4,621	13,184	3.65	
Book	Total	20,102	9,578	141,510	7.03	description, categories, rating, title, review, price
	Train	16,864	9,578	118,638	7.03	
	Valid	1,723	4,957	12,233	7.10	
	Test	1,515	4,919	10,639	7.02	

has been widely used in prior intent-aware methods [39, 67], to encourage different intent embeddings to capture distinct semantic aspects: $\mathcal{L}_{\text{ortho}} = -\frac{1}{|\mathcal{G}|(|\mathcal{G}|-1)} \sum_{i=1}^{|\mathcal{G}|} \sum_{j=1, j \neq i}^{|\mathcal{G}|} \|\hat{\mathbf{e}}_{c_i} - \hat{\mathbf{e}}_{c_j}\|_2^2$, where $\hat{\mathbf{e}}_c$ denotes the L2-normalized embedding of intent c .

Inference. At test time, SPRINT does not invoke LLM to generate intents. Instead, SPRINT leverages the trained intent predictor to identify relevant intents and incorporate them into the prediction process, without incurring additional LLM latency for intent extraction. The next item is predicted as: $\hat{i} = \arg\max_{j \in \mathcal{I}} \sigma(\mathbf{e}_j^\top \tilde{\mathbf{h}}_s)$.

5 Experiments

5.1 Experimental Setup

5.1.1 Datasets. We use three real-world datasets: Beauty (Amazon) [42], Yelp [68], and Books (Amazon) [42], following [33, 36, 39, 44, 47]. For preprocessing, we follow the conventional protocol [6, 36, 44] and apply 5-core filtering. For data splitting, we sort all sessions by the timestamp of their last event and use the 80th and 90th percentiles of these timestamps as cut-off points to split the data into training, validation, and test sets in an 8:1:1 ratio. Table 1 presents detailed statistics.

5.1.2 Baselines. We group the baselines into three categories and compare SPRINT with representative methods from each category.

- (1) **Implicit modeling:** infers intents in an unsupervised manner.
 - **ELCRec** [39] learns intent embeddings via a push-pull loss to enrich representations with the dominant intent.
 - **MiaSRec** [6] dynamically models and selects multiple intents per session to guide item distribution.
- (2) **Explicit modeling:** infers explicit intents from item metadata.
 - **FAPAT** [36] models intents from frequent attribute patterns and then uses them to augment session representations.
 - **CAFe** [28] treats item categories as explicit intents and captures both coarse- and fine-grain sequential dynamics.
- (3) **LLM-enhanced CRM:** uses outputs from LLMs (e.g., intents, text embeddings or profiles) to enhance CRMs.
 - **LLM4SBR** [44] uses an LLM to infer long- and short-term view intents from item titles within each session and incorporates them into the SBR model.
 - **LLM-ESR** [33] converts item-level features into LLM semantic embeddings and fuses this semantic view with collaborative view from the CRM in a dual-view framework.
 - **RLMRec** [47] leverages LLM-derived semantic representations of user/item profiles to enhance CRM learning via two

Table 2: Overall performance comparison. * indicates $p < 0.05$ for the paired t-test against the best baseline.

Method		H@5		N@5		SASRec		H@20		N@20		BERT4Rec		H@20		N@20	
						H@10	N@10					H@10	N@10				
Beauty	Backbone	0.0198	0.0132	0.0297	0.0164	0.0456	0.0203	0.0194	0.0127	0.0310	0.0165	0.0538	0.0221				
	ELCRec	0.0267	0.0170	0.0357	0.0199	0.0590	0.0258	0.0228	0.0149	0.0417	0.0210	0.0650	0.0268				
	MiaSRec	0.0134	0.0073	0.0254	0.0111	0.0650	0.0210	0.0151	0.0086	0.0301	0.0135	0.0714	0.0238				
	FAPAT	0.0250	0.0159	0.0383	0.0201	0.0594	0.0254	0.0241	0.0149	0.0379	0.0194	0.0628	0.0257				
	CaFe	0.0177	0.0108	0.0318	0.0154	0.0521	0.0204	0.0181	0.0112	0.0318	0.0156	0.0589	0.0225				
	LLM4SBR	0.0194	0.0122	0.0353	0.0174	0.0486	0.0208	0.0181	0.0116	0.0319	0.0161	0.0465	0.0198				
	LLM-ESR	0.0168	0.0104	0.0323	0.0154	0.0503	0.0199	0.0284	0.0185	0.0473	0.0244	0.0766	0.0317				
	RLMRec-Con	0.0215	0.0139	0.0349	0.0180	0.0624	0.0249	0.0284	0.0207	0.0447	0.0258	0.0684	0.0318				
	RLMRec-Gen	0.0207	0.0123	0.0357	0.0172	0.0576	0.0227	0.0297	0.0191	0.0413	0.0229	0.0697	0.0299				
	ProfileRec	0.0224	0.0146	0.0357	0.0188	0.0598	0.0250	0.0263	0.0179	0.0413	0.0227	0.0671	0.0291				
	SPRINT	0.0331*	0.0201*	0.0503*	0.0256*	0.0740*	0.0316*	0.0297	0.0211	0.0435	0.0255	0.0680	0.0316				
	SPRINT+LE	0.0293	0.0192	0.0430	0.0236	0.0568	0.0270	0.0349*	0.0244*	0.0503*	0.0293*	0.0817*	0.0372*				
Yelp	Backbone	0.0323	0.0201	0.0554	0.0275	0.0867	0.0354	0.0483	0.0304	0.0831	0.0416	0.1356	0.0548				
	ELCRec	0.0355	0.0227	0.0612	0.0310	0.0985	0.0403	0.0533	0.0336	0.0877	0.0446	0.1338	0.0562				
	MiaSRec	0.0192	0.0112	0.0441	0.0190	0.0913	0.0310	0.0502	0.0275	0.0894	0.0400	0.1474	0.0547				
	FAPAT	0.0328	0.0206	0.0545	0.0275	0.0861	0.0355	0.0558	0.0356	0.0940	0.0478	0.1494	0.0617				
	CaFe	0.0351	0.0226	0.0606	0.0308	0.0974	0.0400	0.0564	0.0360	0.0909	0.0472	0.1409	0.0598				
	LLM4SBR	0.0325	0.0205	0.0552	0.0278	0.0911	0.0368	0.0552	0.0349	0.0925	0.0469	0.1467	0.0605				
	LLM-ESR	0.0363	0.0251	0.0606	0.0329	0.1031	0.0434	0.0542	0.0271	0.0883	0.0447	0.1422	0.0583				
	RLMRec-Con	0.0351	0.0225	0.0571	0.0295	0.0894	0.0377	0.0593	0.0377	0.0955	0.0493	0.1524	0.0636				
	RLMRec-Gen	0.0349	0.0224	0.0552	0.0289	0.0901	0.0376	0.0585	0.0375	0.0967	0.0498	0.1531	0.0640				
	ProfileRec	0.0451	0.0290	0.0765	0.0390	0.1193	0.0498	0.0588	0.0376	0.0961	0.0497	0.1540	0.0642				
	SPRINT	0.0495*	0.0319*	0.0793	0.0414*	0.1272*	0.0535*	0.0600	0.0377	0.1001	0.0506	0.1550	0.0644				
	SPRINT+LE	0.0476	0.0298	0.0798*	0.0402	0.1263	0.0519	0.0628*	0.0404*	0.1037*	0.0535*	0.1577*	0.0671*				
Book	Backbone	0.0165	0.0101	0.0269	0.0135	0.0466	0.0184	0.0511	0.0366	0.0662	0.0414	0.0873	0.0467				
	ELCRec	0.0226	0.0141	0.0367	0.0185	0.0578	0.0239	0.0560	0.0396	0.0751	0.0458	0.0949	0.0507				
	MiaSRec	0.0162	0.0096	0.0309	0.0143	0.0692	0.0239	0.0354	0.0225	0.0665	0.0325	0.1102	0.0435				
	FAPAT	0.0203	0.0129	0.0318	0.0166	0.0488	0.0208	0.0523	0.0364	0.0693	0.0419	0.0915	0.0474				
	CaFe	0.0235	0.0150	0.0384	0.0198	0.0614	0.0257	0.0598	0.0442	0.0743	0.0489	0.0967	0.0545				
	LLM4SBR	0.0164	0.0091	0.0248	0.0119	0.0405	0.0158	0.0578	0.0419	0.0752	0.0476	0.1002	0.0539				
	LLM-ESR	0.0191	0.0126	0.0264	0.0149	0.0380	0.0178	0.0487	0.0343	0.0698	0.0410	0.0973	0.0480				
	RLMRec-Con	0.0224	0.0128	0.0376	0.0176	0.0585	0.0228	0.0550	0.0386	0.0752	0.0451	0.0978	0.0507				
	RLMRec-Gen	0.0210	0.0135	0.0329	0.0173	0.0525	0.0223	0.0636	0.0470	0.0832	0.0534	0.1056	0.0590				
	ProfileRec	0.0293	0.0183	0.0503	0.0250	0.0763	0.0315	0.0672	0.0490	0.0901	0.0565	0.1163	0.0630				
	SPRINT	0.0341*	0.0224*	0.0548*	0.0290*	0.0796*	0.0352*	0.0722	0.0521	0.0936	0.0590	0.1196	0.0655				
	SPRINT+LE	0.0164	0.0110	0.0265	0.0142	0.0378	0.0170	0.0749*	0.0533*	0.0971*	0.0605*	0.1250*	0.0676*				

variants: **RLMRec-Con** projects semantic features into the CRM space, and **RLMRec-Gen** does the reverse.

- **ProfileRec** denotes the method proposed in [38]. It is a state-of-the-art LLM-based profiling method that employs a self-reflection process to generate reliable dual profiles (self and neighbor) for each user and item. These profiles are aligned via contrastive learning to enhance CRMs.

Note that we do not include methods that use LLMs as standalone CRMs [53, 61] in our baselines, as it is a different research direction. Instead, we provide a separate analysis in Table 4. We use GPT-3.5-turbo for all LLM-based methods. Results obtained with Llama-3.3-70B-Instruct are provided in Table 3. For backbone SBR models, we use **SASRec** [18] and **BERT4Rec** [52] as many baselines commonly employ Transformer [54] architectures.

5.1.3 Model variants. We note that all LLM-based baselines (i.e., LLM4SBR, LLM-ESR, RLMRec, and ProfileRec) incorporate LLM embeddings (LE) of item metadata as additional item features. SPRINT is compatible with this design. To ensure a fair comparison, we also report the results of this variant, denoted as **SPRINT+LE**. Specifically, for each item in a session, we obtain a trainable base embedding and a frozen LLM-derived embedding. For a fair comparison under a fixed dimensionality budget, the LLM-derived embedding is projected to $d/2$ dimensions using a two-layer MLP, and combined with the base embedding of the same dimensionality to form a d -dimensional session representation. Each branch is encoded using the same SBR model, and the resulting outputs are concatenated to obtain the final session embedding. For LLM embeddings, we employ OpenAI’s text-embedding-ada-002, except in Table 3 where embeddings from the corresponding Llama model are adopted.

Table 3: Performance comparison of LLM-enhanced CRMs on the Beauty dataset using Llama-3.3-70B-Instruct.

Method		H@5	N@5	H@10	N@10	H@20	N@20
SASRec	LLM4SBR	0.0159	0.0107	0.0293	0.0149	0.0435	0.0185
	LLM-ESR	0.0073	0.0038	0.0099	0.0046	0.0159	0.0061
	RLMRec-Con	0.0219	0.0133	0.0357	0.0176	0.0602	0.0239
	RLMRec-Gen	0.0207	0.0125	0.0301	0.0155	0.0546	0.0217
	ProfileRec	0.0258	0.0166	0.0375	0.0204	0.0620	0.0264
	SPRINT	0.0314*	0.0194*	0.0512*	0.0257*	0.0744*	0.0316*
BERT4Rec	SPRINT+LE	0.0076	0.0038	0.0140	0.0058	0.0226	0.0080
	LLM4SBR	0.0116	0.0070	0.0220	0.0103	0.0447	0.0160
	LLM-ESR	0.0056	0.0028	0.0112	0.0046	0.0194	0.0066
	RLMRec-Con	0.0232	0.0153	0.0340	0.0187	0.0572	0.0245
	RLMRec-Gen	0.0250	0.0169	0.0374	0.0209	0.0568	0.0258
	ProfileRec	0.0245	0.0167	0.0361	0.0205	0.0688	0.0285
SPRINT	SPRINT	0.0258	0.0170	0.0417*	0.0221	0.0735*	0.0301*
	SPRINT+LE	0.0086	0.0065	0.0118	0.0076	0.0183	0.0091

5.1.4 Evaluation metrics. We evaluate recommendation quality using Hit Rate (H) [9] and NDCG (N) [14] at cutoffs {5, 10, 20}, following prior studies [33, 36, 39, 47, 53]. All results are averaged over five independent runs with different random seeds.

5.1.5 Implementation details. We implement all models in PyTorch with CUDA on RTX 3090 GPUs and AMD EPYC 7413 CPUs. Hyperparameters are tuned via grid search on the validation set. The learning rate is chosen from {1e-3, 2e-3, 4e-3}, L_2 regularization from {1e-5, 1e-4, 1e-3, 1e-2}, and the dropout ratio from {0.1, 0.2, 0.3, 0.4}. Transformer backbones use 2 layers, 2 heads, and an embedding dimension of 64. For SPRINT, the number of top- K neighbor sessions is chosen from {5, 10, 15, 20, 25}, and λ_{intent} , λ_{ortho} are chosen from {1e-4, 1e-3, 1e-2, 1e-1}. Baseline hyperparameter ranges follow the original papers.

5.2 Performance Comparison

Table 2 presents the performance comparison. Overall, the proposed framework consistently outperforms all baselines across all datasets and metrics. This improvement stems from three key factors.

First, in contrast to implicit intent modeling (i.e., ELCRec, MiaS-Rec), SPRINT explicitly infers intents via LLM reasoning, providing semantically meaningful guidance to the SBR model. Second, compared to explicit intent modeling (i.e., FAPAT, CaFe) that directly utilize raw item features as pseudo-intents, SPRINT generates semantically rich and nuanced session-level intents by leveraging the reasoning capabilities and world knowledge of LLMs. Third, unlike LLM-enhanced CRMs (i.e., LLM4SBR, LLM-ESR, RLMRec, ProfileRec), SPRINT refines and validates intents based on recommendation performance via a P&C loop coupled with a global intent pool, ensuring highly reliable and informative intents.

Interestingly, SPRINT with LE outperforms SPRINT only when paired with BERT4Rec. We speculate that LLM embeddings help alleviate data sparsity in scenarios where BERT’s bidirectional modeling struggles, such as sparse interactions in Beauty, very short sessions in Yelp (average length < 4), and the large item space in Book. In contrast, incorporating LE into SASRec degrades performance on Book. We attribute this to Book’s relatively longer sequences combined with SASRec’s unidirectional modeling, which makes the model more susceptible to noise introduced by LLM embeddings. This degradation trend is consistently observed in other LE-equipped baselines, including LLM4SBR and LLM-ESR. This highlights the need for further investigation into how the effectiveness of LLM embeddings depends on model architecture and data characteristics, which we leave for future work.

Results with open-source LLM. Table 3 presents the results of LLM-enhanced methods on Beauty using an open-source LLM and its embeddings. Overall, SPRINT significantly outperforms all baselines, showing similar patterns to Table 2. This result highlights the generalizability of our method across various LLMs. Interestingly, models equipped with LE (i.e., LLM-ESR, SPRINT +LE) perform poorly. We attribute this to the representational mismatch between the high-dimensional LLaMA embeddings (8192) and the much lower-dimensional SBR model (64), indicating that more sophisticated alignment techniques may be required.

5.3 Efficiency Analysis

As real-world systems process massive numbers of sessions, generating intents via LLMs for every session is highly inefficient for both training and inference. In this section, we analyze the efficiency of SPRINT. We use SASRec as the backbone SBR model.

Accuracy–efficiency trade-off. Figure 3 shows the trade-off between performance and training/inference time. SPRINT achieves superior accuracy with significantly reduced training time compared to LLM-based user profiling methods, while maintaining inference latency comparable to the LLM-free SASRec. We attribute this favorable balance to two key design choices: (1) informative and reliable intents enhance the SBR model’s accuracy, and (2) selective LLM-based intent generation for uncertain sessions, combined with a lightweight intent predictor, which significantly reduces computational overhead. As a result, SPRINT effectively addresses both uninformative outputs (C1) and scalability limitations (C2).

Table 4: Recommendation accuracy and total inference time of PO4ISR (LLM-as-SBR) vs. SPRINT (LLM-enhanced SBR). Total inference time refers to the cumulative time required to generate predictions for all test sessions.

Dataset	Method	H@5	N@5	H@10	N@10	Inf. Time
Beauty	PO4ISR	0.3996	0.2872	0.5491	0.3351	53.5 min
	SPRINT	0.3372	0.2513	0.4310	0.2814	0.12 s
	SPRINT+LE	0.4293	0.2769	0.6718	0.3550	0.31 s
Yelp	PO4ISR	0.2764	0.1171	0.4877	0.2447	371.5 min
	SPRINT	0.6015	0.4838	0.7158	0.5206	0.23 s
	SPRINT+LE	0.6368	0.5033	0.7600	0.5430	0.57 s
Book	PO4ISR	0.3892	0.3025	0.5124	0.3419	161.9 min
	SPRINT	0.4280	0.3320	0.5608	0.3749	0.14 s
	SPRINT+LE	0.5246	0.3811	0.6972	0.4366	0.39 s

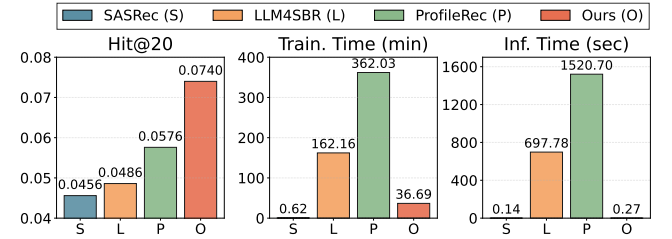


Figure 3: Comparison of performance and efficiency on Beauty. "Train. Time" denotes the total time, including LLM-based user profiling for session contexts (e.g., profiles or intents) and subsequent SBR model training. "Inf. Time" denotes the inference latency for the entire test set.

Comparison with an LLM-as-SBR model. A key advantage of SPRINT is its elimination of LLM dependency at inference time. To highlight this efficiency, we compare SPRINT with PO4ISR [53], a state-of-the-art LLM-as-SBR method that performs both intent generation and recommendation using LLMs during inference. Following [53], we use 20 candidate items, consisting of 19 negative items from a pretrained SASRec and one ground-truth item. Table 4 shows that SPRINT with LE significantly outperforms PO4ISR while drastically reducing inference time. Notably, even without LLM embeddings, SPRINT shows highly competitive performance on all datasets except Beauty. These results highlight the efficiency of SPRINT, which uses only a lightweight intent predictor at test time and requires no LLM inference, enabling scalable deployment.

5.4 Design Choice Analysis

5.4.1 Ablation study. Table 5 presents ablation results on Beauty. Notably, using all proposed components yields the best performance, confirming their efficacy. Specifically, incorporating (A) LLM-generated intents shows a significant improvement compared to when excluding this component. The inclusion of (B) the P&C loop with the GIP further enhances performance, showing its capability to generate reliable and informative intents. Note that the GIP cannot be meaningfully ablated; without it, the unbounded intent space renders intent prediction and fusion in Stage 2 infeasible. Finally, both enrichment variants, (C) self-prediction and (D) neighbor-aggregated prediction, show marked improvements, indicating that our strategy effectively performs intent label enrichment by leveraging both self-confidence and collaborative signals.

Table 5: Ablation study: (A) LLM-generated intents, (B) P&C loop with GIP, (C) self-enrichment, and (D) neighbor-enrichment. (C) and (D) correspond to intent label enrichment using self-prediction and neighbor-prediction only.

(A)	(B)	(C)	(D)	H@5	N@5	H@20	N@20
×	×	×	×	0.0198	0.0132	0.0456	0.0203
✓	✓	×	×	0.0271	0.0164	0.0585	0.0257
✓	✓	✓	×	0.0310	0.0192	0.0619	0.0280
✓	✓	✓	✓	0.0323	0.0195	0.0632	0.0282
✓	✓	✓	✓	0.0327	0.0197	0.0675	0.0295
✓	✓	✓	✓	0.0331	0.0201	0.0740	0.0316

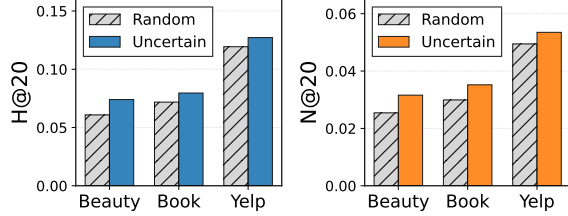


Figure 4: Random vs. uncertainty-aware session selection.

Figure 4 compares the performance of our uncertainty-aware session selection against a random selection. Our approach consistently outperforms the random selection across all datasets. This result confirms that prioritizing high-uncertainty sessions is more effective than random sampling, as it strategically applies LLM calls to the cases where nuanced reasoning is most needed.

5.4.2 Hyperparameter study. We provide analysis to guide the hyperparameter selection of SPRINT. First, we analyze the effect of the intent supervision ratio, i.e., the proportion of sessions for which LLM-generated intents are used. In our main experiments, we generate LLM-based intents only for the top 10% most uncertain sessions (§ 4.1.1). As shown in Figure 5, SPRINT achieves competitive performance even with partial intent supervision and significantly outperforms the 0% setting. We attribute this robustness to the synergy between validated intents from the P&C loop and collaborative intent enrichment, which enables effective propagation of high-quality intent information. Therefore, SPRINT effectively generalizes learned intent patterns to the remaining sessions without explicit LLM intent label supervision.

Figure 6 (left) provides the results of varying λ_{intent} and λ_{ortho} on Yelp. Setting either weight to 0.0 leads to degraded performance, confirming the necessity of both loss terms. Meanwhile, non-zero values consistently improve performance, indicating stable tuning with low sensitivity to exact values. Finally, Figure 6 (right) examines the impact of the number of top- K neighbor sessions (§4.2.2) on Book, where $K = 0$ indicates no neighbor usage. Performance drops sharply at $K = 0$, highlighting the importance of neighbor-based intent enrichment. Performance peaks at $K = 5$, suggesting that a small set of relevant neighbors is sufficient for effective enrichment.

5.5 Case Study

Figure 7 presents a case study. We compare SPRINT with LLM4SBR, a state-of-the-art LLM-enhanced SBR method that explicitly generates interpretable intents. Note that other LLM-based baselines focus on general user profiling in the form of embeddings or personas rather than explicit intent modeling, making direct intent-level comparison via visualization infeasible. Our inferred intents

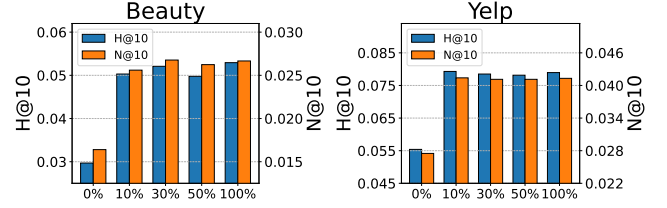


Figure 5: Results with varying amounts of LLM intents.

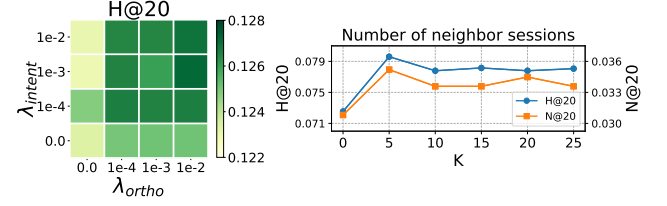


Figure 6: Impact of λ_{intent} , λ_{ortho} , and top- K neighbors.

(e.g., “Fantasy adventure with suspense”, “Emotionally engaging storyline preference”) capture the core characteristics of both the test session and the target item. In contrast, LLM4SBR derives intents directly from item titles (e.g., “Keys to the Demon Prison”, “Deadly Little Secret”), resulting in vaguer and less informative representations. Consequently, our more informative intents lead to improved ranking performance. The t-SNE visualization further shows that the test session embedding is closely aligned with its inferred intents, highlighting the effectiveness of our multi-intent inference process. Overall, this supports that SPRINT improves both accuracy and explainability through high-quality intents.

6 Conclusion

In this paper, we propose SPRINT, a two-stage framework designed to address two unique challenges in LLM-based intent profiling for SBR. For **C1**, SPRINT iteratively refines and validates LLM-generated intents via the P&C loop with the global intent pool, while further enriching them using collaborative signals. For **C2**, SPRINT strategically invokes LLMs via uncertainty-aware session selection and employs a lightweight, LLM-free intent predictor. Our experiments show that SPRINT not only outperforms the baselines but also provides meaningful interpretability. We believe SPRINT provides a practical path toward the scalable and effective deployment of LLM-enhanced SBR systems in real-world scenarios. Future directions include adapting SPRINT to continual settings where user interests evolve over non-stationary data streams [24, 25].

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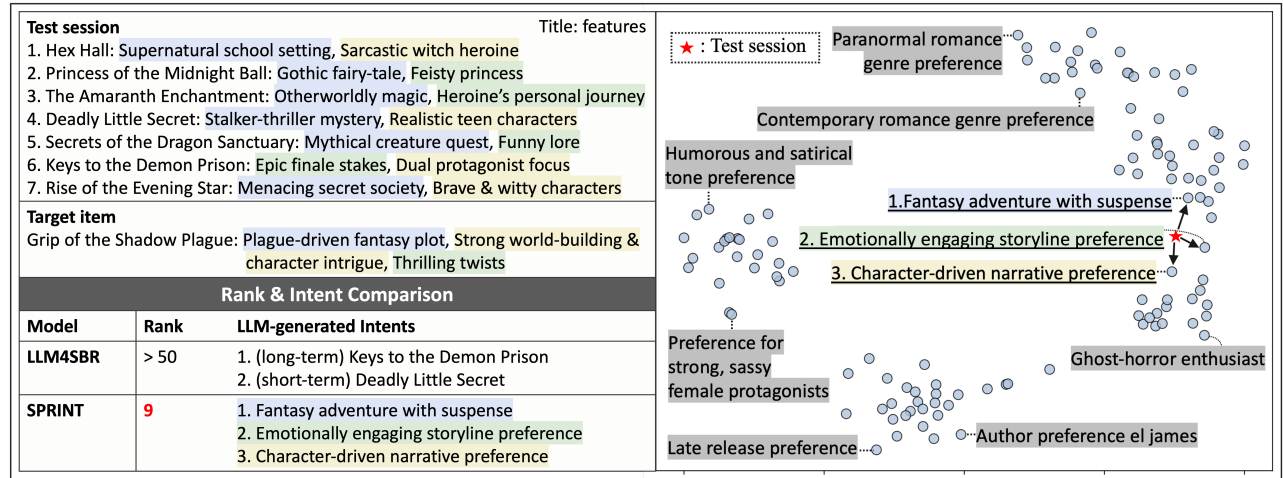


Figure 7: Case study on the Book dataset. (Left) Rank and intent comparison for a test session. Same-colored boxes indicate similar semantic meanings across the test session, target item, and inferred intents. (Right) t-SNE visualization of the learned intent embeddings. The red star denotes the embedding of the test session shown on the left.

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